Towards Intelligent Immersive Healthcare: A Systematic Review on the Integration of AI and XR in Medical Applications

Category: Research

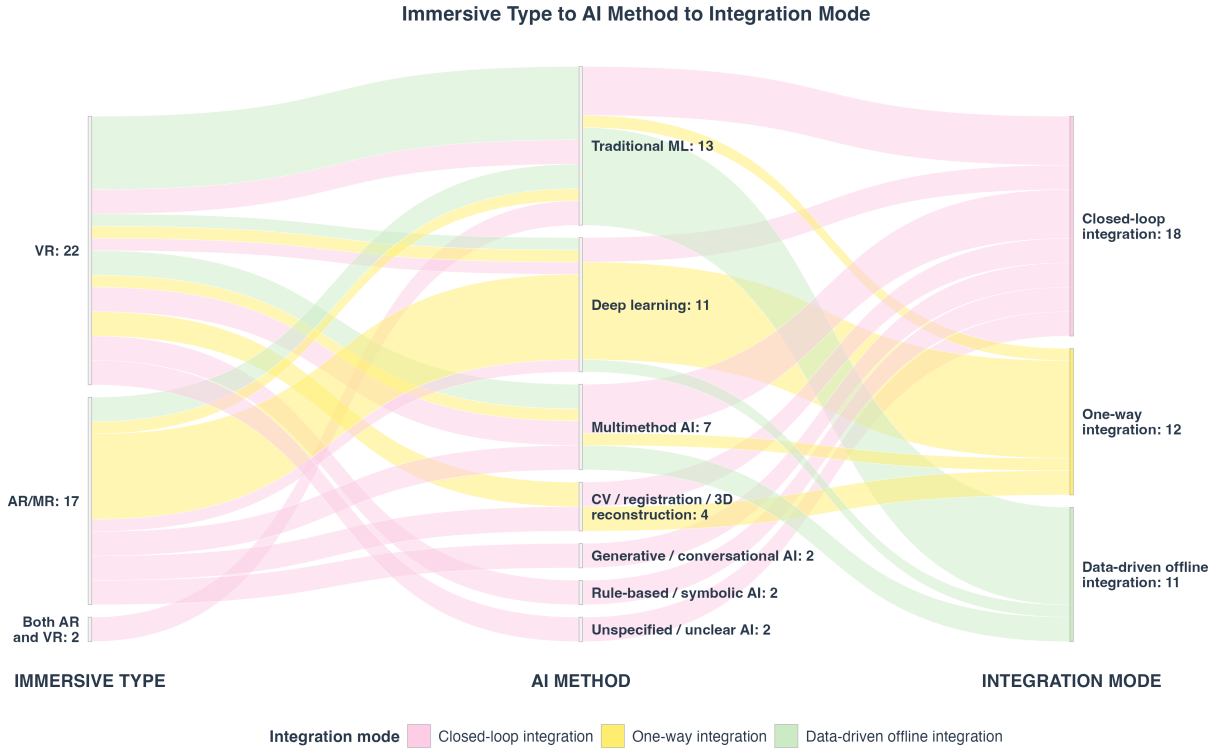


Figure 1: Alluvial mapping of immersive type, AI method family, and integration mode; flow width represents included-study counts.

Abstract

The integration of Artificial Intelligence (AI) and Extended Reality (XR) offers opportunities to develop more adaptive, interactive, and context-aware healthcare systems. However, evidence on how these technologies are integrated into clinically relevant applications remains fragmented, with substantial variation in their roles, technologies, users, and evaluation approaches. This systematic review synthesises 41 clinically relevant empirical studies to characterise AI-XR integration across application domains, users, AI methods, XR technologies, integration approaches, reported outcomes, and clinical implementation maturity. Following PRISMA guidelines, the review systematically identified and analysed how existing studies combine AI and XR across medical applications. The findings show that AI-XR research has progressed beyond proof-of-principle, with systems increasingly evaluated with patients, clinicians, and clinical data across diverse applications. AI-XR applications were concentrated in surgery, mental health, and rehabilitation. Distinct integration modes emerged, with the relationship between AI and XR shaped primarily by the clinical task rather than the AI method or XR hardware. Despite generally positive reported outcomes, evidence remained concentrated in small-scale feasibility and pilot studies, with no study demonstrating large-scale testing or routine clinical adoption. These findings provide a structured view of the emerging AI-XR landscape and highlight the importance of task-specific integration, clinically meaningful interaction, and stronger evidence of implementation in real-world settings.

Index Terms: Artificial Intelligence, Extended Reality, Augmented Reality, Virtual Reality, Mixed Reality, Systematic Review, Clinical Applications, Technology Readiness, Systematic Literature Review, PRISMA.

1 Introduction

Artificial Intelligence (AI) and Extended Reality (XR) are increasingly explored as complementary technologies within healthcare. AI enables computational analysis, pattern recognition, prediction, and adaptive decision support [53, 58, 62]. At the same time, XR, which includes Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), offers spatial, interactive, and embodied modalities for presenting and engaging with information [19, 32, 44, 29]. The integration of these technologies enables the combination of computational intelligence with immersive interaction, potentially enabling healthcare applications that are adaptive, context-aware, and responsive to users and clinical tasks [54].

The potential of AI-XR integration extends beyond simply incorporating two technologies within a single application. AI can enhance XR by recognising users or activities, interpreting clinical or behavioural data, adapting environments and interventions, generating content, or providing personalised feedback [14, 34, 38, 54]. Conversely, XR can offer spatial and interactive interfaces for visualising, interpreting, and acting upon AI-generated information [19, 54, 60]. For instance, AI may analyse movement during an XR-based rehabilitation session and adapt feedback accordingly, or an XR system may present AI-derived clinical information within

the user's spatial context [14, 38, 46]. These complementary capabilities indicate that AI-XR integration can enable novel forms of clinical interaction not achievable by either technology alone.

Despite growing interest, the landscape of AI-XR applications in healthcare remains fragmented. Existing studies encompass a wide range of medical domains, user groups, AI methodologies, XR platforms, and clinical tasks, with substantial variation in how the two technologies interact [54]. AI and XR may operate as separate components or closely integrated within an application, with AI influencing the XR experience, XR facilitating interaction with AI outputs, or both functioning as interdependent elements of a clinical task [54]. Consequently, it remains unclear how AI and XR are currently integrated in practice, whether integration differs across clinical domains, and what capabilities these approaches enable.

The evidence supporting the clinical value and implementation of AI-XR applications is similarly heterogeneous. Studies report technical, usability, behavioural, and clinical outcomes, while applications range from early prototypes and simulated environments to evaluations tested or deployed in clinical practice [28, 34, 64]. As a result, strong technical performance does not necessarily translate to clinical effectiveness or readiness for routine use.

To address these gaps, this study presents a systematic review, conducted in accordance with PRISMA guidelines, of empirical research integrating AI and XR in clinically relevant healthcare applications [50]. The review characterises how AI and XR are integrated, what outcomes have been reported, and how applications are progressing towards clinical implementation, alongside the clinical domains, users, AI methods, and XR technologies involved. The review addresses the following research questions:

- **(RQ1)**: What clinical domains and users are represented in AI-XR applications?
- **(RQ2)**: What AI methods and XR technologies are used?
- **(RQ3)**: How are AI and XR integrated within clinical tasks?
- **(RQ4)**: What outcomes have been reported, and how mature is clinical implementation?
- **(RQ5)**: What trends, limitations, and evidence gaps characterise AI-XR integration in healthcare?

The remainder of the paper is organised as follows. Section 2 introduces the AI, XR, and AI-XR background. Section 3 describes the systematic review methodology. Section 4 presents the findings across applications, users, technologies, integration modes, outcomes, and readiness. Section 5 discusses implications for clinical translation, limitations, and future research.

2 Background

2.1 AI in Healthcare

AI in healthcare is commonly used as a computational method for interpreting clinical data and supporting decision-making, assessment, or feedback. This includes machine learning, deep learning, computer vision, and natural language processing, which enable tasks such as prediction, classification, segmentation, risk modelling, and automated assessment [53, 58, 62]. AI is increasingly applied across the clinical pathway, including medical imaging, surgery, rehabilitation, and patient monitoring. For example, models can detect abnormalities and segment anatomical structures in medical images [41], estimate patient risk [53], analyse surgical workflow and performance [42], and assess movement quality in rehabilitation [15]. The main value of AI in these settings is not to replace clinical judgement and is not limited to diagnosis. AI can reduce repetitive tasks, structure complex information, and provide consistent analysis that supports clinicians and patients [53, 62].

However, technical performance does not necessarily translate into clinical benefit. Issues including limited interpretability, dataset shift, bias, data quality, privacy, and insufficient external validation can affect the reliability and adoption of clinical AI systems [28, 65]. AI outputs must also be delivered within appropriate clinical contexts and workflows to be useful in practice [28]. Opportunities are present for XR to provide an interactive and spatial environment through which AI-generated information can be presented, experienced, or acted upon.

2.2 XR in Healthcare

XR technologies in healthcare support tasks that go beyond conventional screens. Many medical applications involve spatial orientation, repeated practice, controlled exposure, or interaction within a task environment [17, 19, 32, 34]. VR offers a safe, repeatable simulated environment, whereas AR/MR integrates digital content with the physical environment, ranging from information overlays to spatially registered virtual elements [44, 57]. These technologies are valuable because many medical tasks are not only cognitive but also spatial and embodied.

XR has been increasingly applied across different clinical domains. In surgery, it supports training, planning, navigation, and intraoperative guidance [19, 32, 47, 60]. In rehabilitation, XR enables repeated practice, visual feedback, and more engaging exercise [34], while in mental health, VR provides controlled and repeatable environments for exposure and therapeutic interventions [17, 18]. These applications demonstrate how XR takes on different roles depending on the clinical task, from spatial guidance in surgery to feedback in rehabilitation and behavioural observation in mental health.

These task-facing strengths also reveal the limits of current XR systems. Many systems still rely on fixed content, manual setup, or simple visual display, and do not adapt to patient performance, clinical data, or changes in user behaviour. In surgery, issues like registration errors, limited view, latency, and hardware discomfort can reduce usefulness [19, 47, 60]. In rehabilitation and mental health, engagement alone does not guarantee lasting clinical benefit [17, 18, 34]. Some applications work well in a small study but remain difficult to use in routine care [64].

Immersion is therefore only part of the clinical argument. XR often needs data interpretation, personalisation, feedback, and evaluation within real clinical workflows to become clinically useful, areas where AI can add value, but only if the two technologies are integrated in a meaningful way [54].

2.3 AI-XR Integration in Healthcare

Existing reviews have provided useful evidence on AI in medicine and on XR in medicine, but they typically examine the two technologies separately. Reviews of AI in healthcare tend to focus on diagnosis, prediction, segmentation, clinical decision support, or implementation of AI systems [28, 53, 58]. These reviews explain the value and limits of AI, but rarely examine how AI outputs are presented or used within immersive clinical environments. Reviews of XR in medicine, by contrast, focus on surgical training, rehabilitation, mental health, or medical education [17, 19, 32, 34, 60]. They show the value of immersive technologies, but pay less attention to whether the systems are supported by AI-based analysis, prediction, or feedback.

A smaller group of reviews has begun to examine AI together with immersive technologies. Their contribution is useful but still partial. One broad AI-XR systematic review is not specific to medical applications [54], while domain-focused reviews examine areas such as medical education or mental health rather than the full cross-domain clinical evidence base [11, 61]. Others focus on a single immersive technology, such as VR, without covering AR or MR. Current reviews therefore do not yet provide a clear cross-domain picture of how AI and XR are integrated across clinically relevant medical applications.

Less well synthesised is the relationship between the two technologies once they appear in the same medical application. It is unclear whether AI and XR are simply used side by side or whether they form distinct technical and clinical relationships, and how such patterns vary across medical domains with different task structures. Existing reviews have also rarely examined reported effects together with evidence maturity, study design, sample size, and implementation context [64]. As a result, it is difficult to determine if current AI-XR applications have a real-world clinical impact.

The remaining gap is therefore not whether AI and XR appear in the same studies, but how they are integrated within clinically relevant tasks. A useful synthesis needs to link application domain, user group, AI function, immersive technology, integration relationship, reported outcomes, and evidence maturity. In this review, integration is defined as a functional connection between AI and XR within the same clinical task, system, workflow, or data pathway, such that the two components serve a common clinical objective. This review addresses the identified gap by examining clinical and clinically relevant empirical studies of AI-XR in medical applications under this definition.

3 Methodology

3.1 Research design

This study aimed to systematically synthesise empirical evidence on the integration of AI and XR, including AR, VR, and MR, in clinically relevant medical applications. The review was conducted and reported in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines [50] and focused on studies in which AI and XR were functionally connected within the same clinical or clinically relevant task, system, workflow, or data pathway. The review examined how these technologies are integrated and applied, the outcomes reported, and their progression towards clinical implementation. Given the heterogeneity of the included studies, findings were synthesised using narrative synthesis.

3.2 Search strategy

The literature search was conducted on 24 February 2026 and covered PubMed[1], Scopus[2], and Web of Science[3]. The publication date range was limited to 1 January 2020 to 31 December 2025 to focus on recent clinically relevant evidence in a rapidly developing field.

The search strategy revolves around four concepts: (1) immersive technology, (2) AI, (3) medical or healthcare, and (4) empirical clinical studies. The search term used is as follows:

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("Augmented Reality" OR "Virtual Reality"
OR "Mixed Reality" OR "Extended Reality") AND
("Artificial Intelligence" OR "Machine Learning"
OR "Deep Learning" OR "Computer Vision") AND
(healthcare OR "health care" OR patient* OR
clinic* OR medic*) AND ("clinical trial" OR
"randomized controlled trial" OR "randomised
controlled trial" OR "prospective trial" OR
"prospective study" OR "controlled clinical trial"
OR "single-arm interventional trial" OR "pilot
clinical trial")
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Database-specific syntax, including MeSH key terms, was used to preserve the same conceptual logic while respecting different indexing systems. The clinical study terms were used to improve the retrieval of clinically relevant empirical research and did not restrict eligibility to formal clinical trials.

[1]PubMed pubmed.ncbi.nlm.nih.gov/
[2]Scopus: www.scopus.com/pages/home
[3]Web of Science: webofscience.com

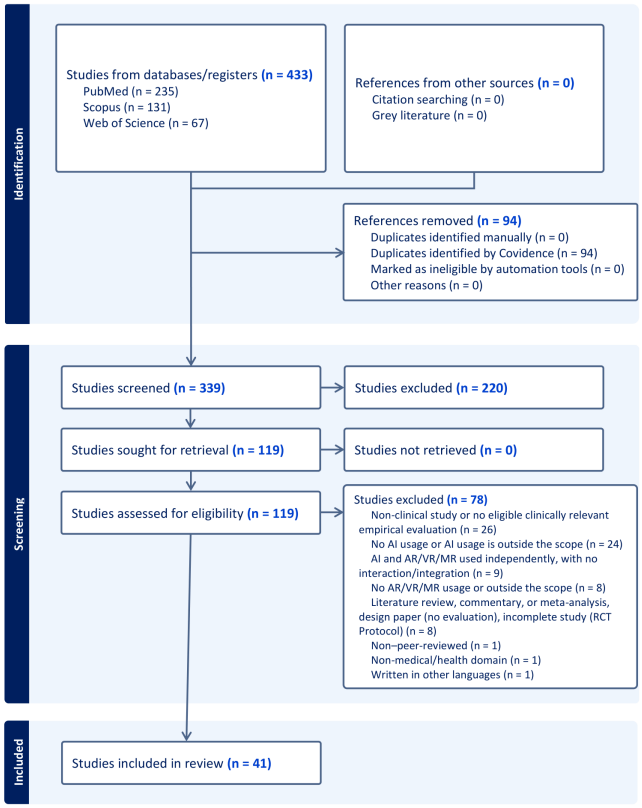


Figure 2: PRISMA flow diagram for study selection.

3.3 Eligibility criteria

Eligibility criteria were predefined and summarised in Table 1. They were refined through discussions with four domain experts to clarify the scope of AI-XR integration, clinically relevant applications, and suitable empirical evidence. Eligible studies had to be peer-reviewed, written in English, published between 2020 and 2025, involve empirical studies in a medical or healthcare domain, involve human participants or data, and functionally integrate AI with XR within the same clinically relevant task, system, workflow, or data pathway. Literature review articles were excluded.

3.4 Screening process

All identified records were imported into Covidence[4] for screening and review management. A total of 433 records were identified from PubMed (n = 235), Scopus (n = 131), and Web of Science (n = 67). No additional records were identified from other sources, citation searching, or grey literature. Before screening, 94 duplicate records were identified and removed, leaving 339 records for title and abstract screening. Two authors independently screened titles and abstracts and subsequently assessed potentially eligible studies at full text against the predefined eligibility criteria. Disagreements were resolved by a third author. At title and abstract screening, 220 records were excluded. Of the 119 studies assessed at full text, 78 were excluded, resulting in 41 studies included in the final review. The detailed process is reported in Figure 2.

3.5 Data Extraction and Synthesis

Data were extracted from the included studies, according to application domain (**RQ1**), target users (**RQ1**), AI methods (**RQ2**), XR technologies (**RQ2**), AI-XR integration (**RQ3**), reported outcomes (**RQ4**), clinical implementation maturity (**RQ4**), and study

[4]Covidence: www.covidence.org/

Inclusion Criteria	Exclusion Criteria
Integration of AI with XR (AR, VR, or MR)	AI or XR absent, out of scope, or used independently with no interaction/integration
Medical / healthcare domain	Non-medical / non-healthcare domain
Clinical or clinically relevant empirical study	Non-clinical study or no eligible clinically relevant empirical evaluation
Human participants, or human participant data	Animal, cadaver, phantom, simulation-only studies, or no human participant data
Empirical study	Literature review, commentary, meta-analysis, design paper without evaluation, protocol, or incomplete study
Written in English	Non-English study
Peer-reviewed	Non-peer-reviewed

Table 1: Inclusion and exclusion criteria.

limitations (**RQ5**). Extracted information was thematically organised and compared to identify common characteristics and patterns in AI-XR applications and their integration within clinical tasks. Clinical implementation maturity was assessed using Technology Readiness Levels (TRLs), informed by health-sector frameworks spanning medical devices, digital technologies, and clinical pathways [21, 39]. Given the heterogeneity of the evidence, findings were synthesised narratively, supported by descriptive statistics and charts. The data extraction coding is provided in the Supplementary Material.

4 Results

4.1 Study Overview

The search and screening process produced 41 included clinical and clinically relevant empirical studies. Publication activity concentrated toward the end of the review period: two studies appeared in 2020, six studies were published in each year from 2021 to 2024, and 15 studies appeared in 2025.

The 2025 increase was not driven by a single area but was distributed across surgery, rehabilitation, mental health, supportive care, neurology, and public health. Surgery was consistently represented throughout and contributed substantially to the 2025 rise. Rehabilitation showed the most visible recent expansion, while mental health appeared across multiple years without a steady increase. Supportive-care applications emerged mainly in the later years, whereas public health remained an isolated example.

4.2 Application Domains and Target Users

To address **RQ1**, we examined the medical application domains in which AI and XR were used together and their target users.

4.2.1 Application Domains

As shown in Figure 3, the most evaluated domains were *surgery* (n = 13), *mental health* (n = 10), and *rehabilitation* (n = 9). *Supportive and care-related* applications accounted for four studies, followed by *neurology* (n = 3), and *diagnostics/pathology* and *public health* for one study each. AI-XR research spans a range of clinical healthcare domains, yet the evidence was unevenly distributed.

In *surgery*, AI-XR applications were shaped by spatial tasks, including 3D reconstruction, registration, planning, navigation, and intraoperative overlay. Examples included automatic registration for laparoscopic liver surgery [31], AI-supported VR planning for lung segmentectomy [56], AR navigation for lumbar discectomy [24], and AI-AR guidance for robotic prostatectomy [52]. Surgery is a natural fit as AI segments anatomy or registers models while XR visualises that spatial information during planning and guidance.

Mental-health applications centred on immersive behavioural tasks paired with AI-based prediction, classification, or conversational support. For example, VR-based stabilisation for post-traumatic stress symptoms was combined with machine-learning prediction of symptom change [59]. Other studies addressed social anxiety, spider phobia, and treatment-resistant schizophrenia through VR exposure, serious games, or avatar therapy, with AI used for prediction, adaptive exposure, or analysis of therapy transcripts [7, 25, 26, 36, 45, 51]. The mental health domain used immersive environments not as visualisation tools but as contexts for eliciting behaviour, speech, physiology, or interaction data.

Rehabilitation formed a third major cluster, linking AI-XR systems to performance monitoring, motor control, and adaptive feedback. Examples included AI-based assessment within VR hand-exoskeleton training for post-stroke rehabilitation [14], machine-learning-supported phantom motor execution with AR/VR feedback [38], and myoelectric VR serious gaming for chronic stroke [46]. Rehabilitation suits closed feedback loops because movement, muscle activity, and task performance can be captured and used to adjust therapy in real time.

The remaining domains were smaller and covered applications related to care support, monitoring, assessment, and behaviour-change intervention. *Supportive and care-related* applications included VR-AI support for hot flashes and psychological wellbeing in women with breast or ovarian cancer [22], AI-assisted VR preparation for paediatric endoscopy [8], digital wound-area measurement using augmented reality and K-means-based image analysis [10], and AR-supported quality-of-life monitoring for colostomy care [67]. These studies used AI-XR less for procedural precision than for patient preparation, symptom support, measurement, and ongoing care interaction.

Neurology studies focused mainly on cognitive or functional assessment and support. Examples included an immersive virtual supermarket with machine-learning-based assessment for early neurocognitive impairment [63], digital biomarker analysis in Alzheimer's disease management [20], and an immersive VR application for Alzheimer's and dementia support [1]. In this domain, XR was used to create structured task environments, while AI supported assessment, monitoring, or personalised support.

Diagnostics/pathology was represented by an AI-empowered microscope designed to improve Ki67 assessment concordance in pathology [9], with AI-XR integration supporting clinician-facing diagnostic assessment.

Public health was represented by an adaptive AI-based VR sports intervention for adolescents with excess body weight [66], using personalised virtual coaching and immersive exercise to support behaviour change.

4.2.2 Target Users

Target-user coding showed a predominantly *patient- and clinician-centred* evidence base. *Patients* were the intended users in 25 studies, *clinicians* in 13, and *mixed users* in three (Figure 3). Clinician-facing studies concentrated in surgery and diagnostic-support workflows, while patient-facing studies dominated mental health, rehabilitation, supportive care, and public health.

4.3 Technical Characteristics of AI-XR Systems

To address **RQ2**, we characterised the AI methods and functions, XR modalities, and hardware used across the included studies.

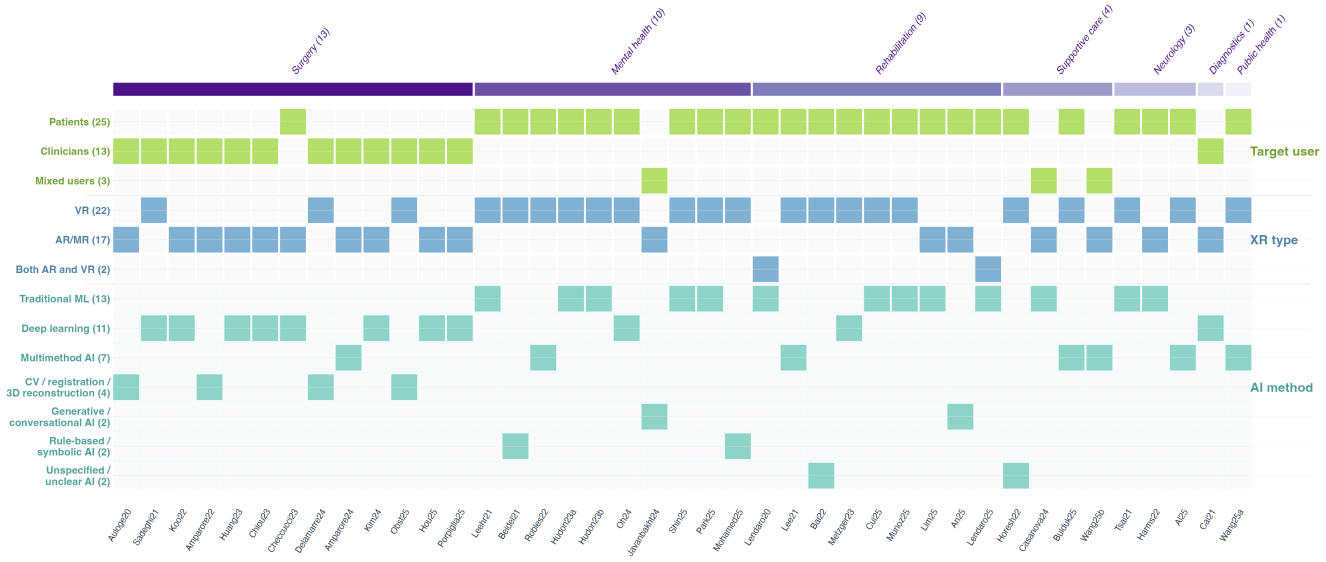


Figure 3: Distribution of application domains, target users, XR types, and AI methods across included studies.

4.3.1 AI Methods and Functions

As shown in Table 2, AI was coded at two levels: *method* and *function*. The first level describes the technical method, such as traditional machine learning, deep learning, multimethod AI, or generative/conversational AI. The second level describes the function of AI within the AI-XR system, such as planning support [23, 56], classification [49, 55], prediction [51, 59], coaching [66], or adaptive control [14], that contributes towards clinical workflow.

Traditional machine learning and *deep learning* were the most frequently reported AI method groups, appearing in 13 studies and 11 studies, respectively (Figure 3). *Traditional machine learning* was used for tasks including predicting the response after VR-based stabilisation for posttraumatic stress symptoms [59] and classifying sarcopenia from Kinect-based MR exercise data [40]. *Deep learning* supported applications including AI-empowered Ki67 pathology assessment and speech decoding with avatar control [9, 43]. *Generative/conversational AI* and *rule-based/symbolic AI* were smaller but relevant categories, appearing in AR-based mental-health support, MR exercise consultation, adaptive exposure, and serious-game therapy [4, 7, 27, 45].

AI function was more directly linked to the clinical domain of each system. In surgery, AI most often supported navigation, planning, registration, or overlay, as seen in AR navigation for lumbar discectomy, AR-guided robotic prostatectomy, and multimodal AR planning for extracerebral tumours [23, 24, 52]. In rehabilitation, AI was used for adaptive feedback, motor control, or performance assessment, including hand-exoskeleton training, phantom motor execution, and myoelectric VR serious gaming [14, 38, 46]. In mental health, AI supported prediction, classification, conversational/coaching support, or behavioural data analysis around VR exposure, stabilisation, and avatar therapy [25, 26, 36, 51, 59]. In supportive care, AI-XR systems were used for preparation, monitoring, guidance, or care interaction, including paediatric endoscopy preparation and colostomy-care monitoring [8, 67].

4.3.2 XR Modalities and Hardware

VR was the most frequently used XR modality, appearing in 22 studies, followed by AR/MR in 17 studies. Two studies used both AR and VR (Figure 3). Examined by domain (Figure 3), a task-related pattern emerged. VR predominated where the task required an immersive environment, such as exposure therapy, social or avatar interaction, rehabilitation training, and behavioural intervention, whereas AR/MR concentrated where the task involved spatial overlay, registration, or guidance, especially surgery and procedural support. XR modality thus tracked the clinical task: VR to create an immersive environment, AR/MR to place digital information into physical or procedural space.

The type of XR hardware used in the included studies was related to both XR modality and clinical setting. VR studies predominantly used commercial head-mounted displays, especially the HTC Vive/Vive Pro Eye and Oculus/Meta Quest or Rift series [1, 48, 49, 55, 59, 63, 66]. A smaller number used older headsets, smartphone-based VR, screen-based VR, serious-game simulators, or custom rehabilitation setups [6, 7, 16, 22, 35, 36, 43, 46]. AR/MR studies showed greater hardware diversity and were more often tied to clinical or procedural use. These included surgical navigation and guidance systems [24, 31, 52], robot-assisted AR platforms with fluorescence-imaging overlay [2, 3], mobile- or tablet-based AR systems [10, 13, 20, 23, 30], and projector- or camera-based AR systems [5, 67]. Only one AR study used wearable AR headsets, including Microsoft HoloLens and Apple Vision Pro [27]. Only two studies used both AR and VR, both in rehabilitation-related applications, including an AR/VR rehabilitation system and a screen-based AR/VR system [37, 38]. Table 3 summarises the XR type and device category for the included studies.

4.4 Patterns of AI-XR Integration Across Clinical Domains

To address **RQ3**, the functional integration of AI and XR within the clinical tasks, systems, workflows, or data pathways of the included studies was examined. Three integration modes emerged based on the direction of information flow, the timing of AI analysis, and the way AI and XR worked together within the clinical task:

- *One-way integration*: systems in which AI outputs flow into the XR display, but user actions within XR do not feed back to change the AI component [23, 56]. AI may generate outputs such as segmentation, reconstruction, registration, prediction, planning, or guidance, and XR presents or uses these outputs through an immersive or spatial interface.
- *Data-driven offline integration*: systems in which XR collects or structures behaviour, image, speech, physiological, clinical, or performance data, and AI then performs diagnosis, classification, prediction, assessment, or modelling after or

AI methods	AI functions	Studies
Traditional machine learning	Classification / diagnosis / screening	[20, 40, 51, 63]
	Prediction / prognosis	[25, 36, 59]
	Adaptive personalisation / closed-loop control	[14, 37, 38, 46]
	Performance or functional assessment	[10, 26]
Deep learning	Classification / diagnosis / screening	[9, 49]
	Segmentation / reconstruction / registration	[31]
	Navigation / planning support	[12, 13, 23, 24, 30, 52, 56]
	Adaptive personalisation / closed-loop control	[43]
Computer vision / image registration / 3D reconstruction	Segmentation / reconstruction / registration	[48]
	Navigation / planning support	[2, 5, 16]
Rule-based or symbolic AI	Adaptive personalisation / closed-loop control	[45]
	Coaching / tutoring / conversational support	[7]
Generative or conversational AI	Coaching / tutoring / conversational support	[4, 27]
Multimethod AI	Classification / diagnosis / screening	[55]
	Navigation / planning support	[3]
	Adaptive personalisation / closed-loop control	[1, 67]
	Coaching / tutoring / conversational support	[66]
	Performance or functional assessment	[35]
	Generative content or guidance	[8]
Unspecified or insufficiently described AI	Adaptive personalisation / closed-loop control	[6, 22]

Table 2: AI method groups, system functions, and constituent studies.

XR type	Device category	Studies
VR	Head-mounted display (HMD)	[1, 8, 14, 22, 25, 36, 45, 48, 49, 51, 55, 56, 59, 63, 66]
	Screen-based VR	[7, 16, 43, 46]
	VR rehabilitation platform	[6, 35]
	Not clearly reported	[26]
AR/MR	Mobile or tablet-based AR	[10, 13, 20, 23, 30]
	AR surgical navigation/guidance system	[24, 31, 52]
	Projector/camera-based AR system	[5, 67]
	Robot-assisted AR platform	[2, 3, 12]
	Head-mounted display (HMD)	[4, 27]
	AR microscope platform	[9]
	Sensor-based mixed-reality system	[40]
AR & VR	AR/VR rehabilitation system	[38]
	Screen-based AR/VR	[37]

Table 3: XR hardware by immersive type and device category.

outside the immersive session [36, 51, 55, 63]. The AI results are not fed back into the XR environment in real time.

- *Closed-loop integration*: systems in which XR provides an adaptive training or interaction environment, users or patients perform tasks, AI analyses their behaviour, performance, or physiological signals, which is then updated accordingly in the VR/AR scene [14, 38, 43, 46].

Closed-loop integration was the most common mode (n = 18), followed by *one-way integration* (n = 12) and *data-driven offline integration* (n = 11). This distribution indicates that AI-XR systems frequently incorporated interaction between the two components, although the nature and timing of this interaction varied.

Figure 1 places immersive type, AI method, and integration mode along a single pathway to examine how they co-occur. VR connected to several AI methods, including traditional machine learning, deep learning, and multimethod AI, while AR linked more often to vision-oriented methods such as deep learning and computer vision/registration. The same AI method, however, appeared across different integration modes, so integration mode was not determined by the AI method but by how each system connected AI and XR within its clinical task. The organising axis is therefore the structure of the clinical task, not the underlying technology stack.

Clinical tasks differ across domains, as does the way AI and XR are combined (Figure 4) and the examples below.

- *Surgery*: One-way integration was predominant, with AI generating or registering spatial information that AR or VR presented for planning or navigation, as in laparoscopic liver registration [31], lung segmentectomy planning [56], lumbar discectomy navigation [24], and multimodal AR planning for extracerebral tumours [23]. Some surgical systems were coded as closed-loop when registration or overlay updated dynamically, such as ICG-driven AR robotic partial nephrectomy [2] and AR/AI vertebroplasty navigation [5].
- *Rehabilitation*: Showed a different architecture, strongly associated with closed-loop systems in which AI contributed to assessment, motor-intent decoding, assistance-as-needed, or adaptive feedback. Examples include VR hand-exoskeleton rehabilitation with real-time assessment and adaptive task difficulty [14], home-based phantom motor execution with machine-learning decoding [38], and myoelectric VR serious gaming for chronic stroke [46].
- *Mental health*: Data-driven offline integration was more common, with XR used to generate behavioural, speech, or physiological data analysed offline, as in posttraumatic-stress response prediction [59], anxiety prediction from VR simulated stimuli [51], and avatar-therapy transcript analysis [25, 26].
- *Other domains*: Supportive and care-related applications, neurology, diagnostics and pathology, and public health showed heterogeneous integration patterns, reflecting the smaller number of studies and differences in clinical objectives.

The cross-domain differences show that AI-XR integration takes different forms depending on the clinical task. The maturity of evidence supporting these integration patterns is examined next.

4.5 Reported Outcomes and Technology Readiness

To address **RQ4**, the reported outcomes and technology readiness of the included AI-XR systems were examined.

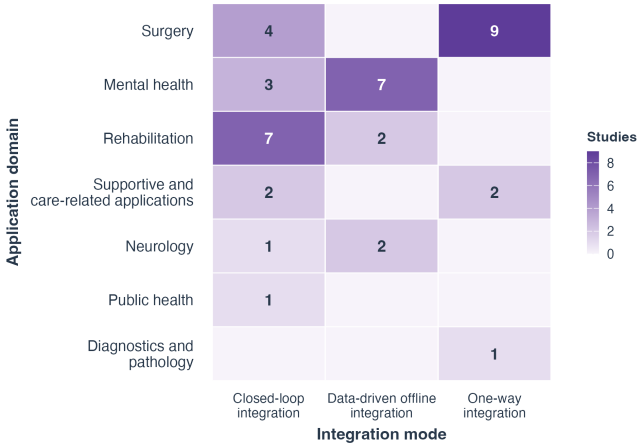


Figure 4: Application domain by integration mode.

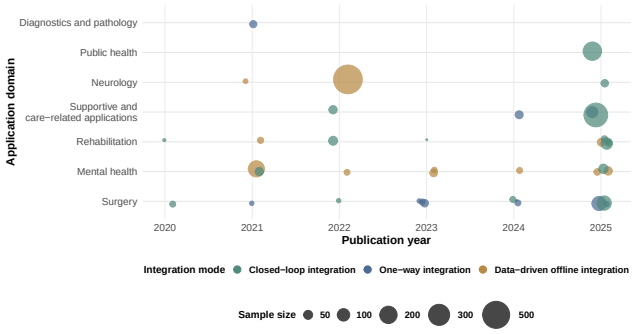


Figure 5: Evidence map by year, application domain, integration mode (bubble colour), and sample size (bubble size, comparable human or patient-derived counts where extractable).

4.5.1 Reported Outcomes

Reported outcome direction was predominantly positive, with 39 studies coded positive, one mixed, and one limited or inconclusive. The positive findings spanned technical performance, feasibility or usability, symptom or behavioural change, clinical-workflow measures, and classification or prediction performance.

This overall positive direction should, however, be interpreted cautiously, as the coding reflects the direction of reported findings rather than their magnitude or strength. Several positive studies reported mixed details beneath the overall label. For example, the AR lumbar discectomy navigation study reported fewer puncture attempts and fluoroscopy shots but no significant difference in operation time or several clinical scores [24]. The AI-3D-AR prostatectomy trial reported lower positive-margin rates and other favourable outcomes, but within one surgical context and design [52]. The XR phantom-limb-pain trial showed improvement in both treatment arms, but phantom motor execution was not superior to phantom motor imagery, hence its mixed coding [37]. The spider-phobia study showed strong group-level benefit from one-session VR exposure but limited, non-significant longer-term prediction, explaining the limited/inconclusive coding of its AI component [36].

4.5.2 Technology Readiness

Given the predominantly positive findings, technology and evidence maturity provided a more informative view of the current state of AI-XR research. TRLs were used to characterise the development and evaluation stage of each system, drawing on established health, care, digital, and pathway innovation frameworks [21, 39]. TRLs assignment considered the extent of human-participant evaluation, scale of testing, and clinical or operational setting demonstrated in each study. TRL was interpreted as an indicator of development and evaluation maturity rather than clinical effectiveness or routine adoption [21, 39, 64].

As shown in Table 4, most studies were classified at TRL 6 (n = 27), followed by TRL 7 (n = 11) and TRL 5 (n = 3). No study reached the higher stages associated with large-scale evaluation, cost-effectiveness assessment, or routine adoption into care pathways. Although some studies involved larger clinical cohorts or controlled designs, higher TRL did not necessarily indicate established clinical effectiveness or operational deployment. Comparatively strong evaluations such as the adaptive VR sports trial in adolescents with excess body weight [66], colostomy-care monitoring with AR and deep learning [67], and AI-3D-AR robotic prostatectomy [52] provided stronger clinical-evaluation evidence than small pilots without demonstrating routine deployment. Cross-tabulating integration mode against TRL (Table 4) shows that this maturity was not evenly shared across architectures. Data-driven offline integration did not reach TRL 7, remaining at TRL 5 and 6 only, while one-way integration (5 of 12 studies) and closed-loop integration (6 of 18) both progressed to larger-cohort clinical testing. Evaluation maturity thus mirrored the integration architecture: offline-modelling systems, concentrated in mental health, have so far been evaluated mainly at the feasibility stage. Surgical one-way guidance and rehabilitation closed-loop systems more often advanced to larger clinical cohorts, although none progressed beyond TRL 7.

These readiness differences can be read alongside the timing, application domain, and scale of each study in a combined evidence map (Figure 5), which plots every included study by publication year and application domain, with bubble colour indicating integration mode and bubble size indicating the comparable human, clinical-user, or patient-derived sample available for evaluation. The map includes 39 studies with comparable sample-size values and excludes two with non-comparable units. Larger evaluations tended to use controlled or randomised designs, including the adolescent VR sports trial [66], colostomy monitoring [67], extracerebral tumour planning [23], and AI-3D-AR robotic prostatectomy [52]. Several technically sophisticated systems, by contrast, were tested only in small feasibility or case-based samples, such as the speech neuroprosthesis with avatar control [43], lung segmentectomy planning [56], and home-based phantom motor execution [38]. Larger samples strengthened the evidence but were not sufficient on their own to establish maturity or general clinical effectiveness. Overall, the predominantly positive results rest on an evidence base concentrated at real-setting piloting, unevenly scaled, and not yet supported by routine-deployment evidence.

5 Discussion

5.1 Principal Findings

This review examined where and how the integration of AI with XR is approaching real-world impact in clinically relevant healthcare applications. Three findings stand out. First, AI-XR integration is no longer speculative: the 41 included studies show that these systems are being evaluated with patients, clinicians, and clinical data across a widening range of medical tasks and domains, with publication activity increasing sharply in 2025. Surgery, mental health, and rehabilitation accounted for the largest share of applications, although AI-XR was also explored across several other areas.

Second, integration patterns are shaped more by clinical tasks than by the specific AI method or XR hardware. Three modes of integration (one-way, data-driven offline, and closed-loop) were identified, with their use varying across clinical domains. Hence, the broad technology label of AI-XR covers substantially different forms of technical and clinical connection, reflecting differences in how AI and XR contribute to particular clinical tasks.

TRL	Definition in a health/digital health context	Closed-loop integration	One-way integration	Data-driven offline integration	Total
TRL 5	Validated in a relevant environment using human participant-related data, but not directly evaluated with human participants [39].	1	1	1	3
TRL 6	A near-final prototype is tested in a small cohort of human participants for performance, usability, safety, characterisation, and reproducibility [39].	11	6	10	27
TRL 7	The system is tested in a larger cohort of human participants, often in a realistic clinical or operational setting. Usability, safety, and performance are closely monitored [39].	6	5	0	11
Total	–	**18**	**12**	**11**	**41**

Table 4: AI-XR integration mode by Technology Readiness Level (TRL).

Third, research activity should not be mistaken for clinical maturity. Although most studies reported almost uniformly positive outcomes, the evidence remained concentrated in small-scale piloting, heterogeneity prevented meta-analysis, and no study reached testing at scale or routine adoption. Taken together, the findings indicate that AI-XR has progressed towards clinically relevant evaluation, but its real-world clinical impact remains to be established.

5.2 Task-Driven Integration

A key finding of this review is the separation of one-way, data-driven offline and closed-loop integration, and the finding that these modes followed the clinical task. Surgery predominantly used one-way integration, in which AI reconstructed, segmented or registered anatomy, and then AR or VR presented it for planning and navigation [23, 31, 56]. A few surgical systems closed the loop, but only where the overlay updated during the procedure itself [2, 3]. Rehabilitation was dominated by closed-loop designs, because movement, muscle activity and task performance can be sensed continuously and used to adapt the task as the patient works [14, 38, 46]. Mental-health studies more often used the immersive environment to elicit behavioural, physiological, or linguistic data, which AI then analysed after the session [26, 36, 59]. The alluvial analysis (Figure 1) made the point clearer. The same AI method family appeared across several integration modes, so the architecture of an AI-XR system reflects the logic of the clinical problem more than the algorithm chosen to address it. This pattern is established only in the domains with sufficient studies: surgery, rehabilitation, and mental health. The single studies in diagnostics and public health domains were insufficient to support any claim about their integration mode.

The predominance of the one-way integration mode in surgery is worth noting. Intraoperative decisions carry immediate clinical risk, and a system that allowed an AI model to alter guidance autonomously would face safety and regulatory demands that few current tools can meet. The observed pattern, in which AI prepares or registers information for subsequent interpretation by the surgeon, may therefore represent an appropriate form of human-AI collaboration rather than technological immaturity. Whether greater integration or closed-loop control is desirable in surgery should consequently be determined by the specific clinical task, risk, and degree of human oversight required.

These findings suggest that AI-XR should not be treated as a single technological category. The umbrella term AI-XR covers systems that differ by clinical tasks, from guiding a surgeon's hands [5, 23, 52], to scoring a stroke patient's movement [35], to modelling a phobia from session data [51]. Treating them as one category obscures more than it clarifies. Prior reviews either examined AI and immersive technology separately, or studied their combination outside the medical scope [54] or within a sub-area such as mental health [61] or healthcare education [11], and could not surface this structure. Organising the evidence by integration mode and clinical task therefore provides a more meaningful basis for comparing AI-XR systems and identifying where deeper integration may be clinically appropriate.

5.3 From Positive Signals to Clinical Impact

The overarching motivation of our review (spanning all **RQ**s) was to determine the extent of clinical impact of AR-XR applications. The evidence is clinically oriented but still some distance from demonstrated real-world impact. The systems were predominantly patient-facing and clinician-facing, and 39 of 41 studies reported positive results. That near-uniformity warrants caution rather than confidence. It is consistent with an early-stage literature dominated by feasibility work, and with the familiar tendency to publish favourable findings, and it sits awkwardly beside the maturity data. Several studies coded as positive also reported important limitations, including non-significant differences on secondary clinical outcomes [24], no advantage over a simpler comparator [37], and weak longer-term prediction [36]. Such nuances can be obscured when the overall direction of findings is interpreted as evidence of clinical benefit.

As observed in clinical AI more broadly [28, 65], strong technical performance and promising pilots results do not necessarily demonstrate that a system improves care across real workflows, varied patient populations, and sustained use. For AI-XR, this gap is further compounded by hardware dependence, cost, user discomfort, and the difficulty of fitting real-time systems into clinical routines. Current evidence therefore reports promise and feasibility far more often than demonstrated clinical effectiveness.

The evidence map in Figure 5 also highlighted limitations in evaluation scale. Larger evaluations were concentrated in a small number of controlled or randomised studies [52, 66, 67], while several technically ambitious systems, including the speech neuroprosthesis with avatar control [43] and home-based phantom motor execution [38], were evaluated in very small samples. Small samples are reasonable at the proof-of-concept stage, but they limit the ability to make solid claims. This pattern is consistent with the technology-readiness findings, in which most systems were situated at TRL 6, with fewer reaching TRL 7 and none demonstrating large-scale testing or routine adoption.

5.4 Ethical, Privacy, Safety and Accountability Considerations

Integrating AI with immersive technology raises concerns that can be more complex than those associated with either technology alone. Immersive systems collect rich and potentially sensitive data, including gaze, movement, voice, physiological responses and behaviour during exposure or therapy tasks, and AI converts these into clinical inferences [33]. The resulting record is more revealing than a conventional one, yet questions of consent, storage and secondary use received little attention in the included studies. These concerns become more pronounced in closed-loop systems. When

an AI component changes what a patient sees, or alters a surgical overlay in real time, responsibility for an error may be distributed across the AI model, software, hardware, clinician, and healthcare organisation, making accountability more complex. Interpretability and trust, already important considerations in clinical AI [28], grow more complicated when outputs are embedded in an immersive scene that can feel seamless and therefore authoritative.

Accessibility is another concern. Reliance on headsets and specialised hardware could widen disparities, although the same can also narrow them, as in the use of VR to bring surgical expertise to a low-income setting [48]. Systems that are at once software and medical devices sit awkwardly within existing categories, and the absence of any near-deployment study suggests that safety monitoring and accountability arrangements for such systems remain underdeveloped. The immersive setting introduces additional safety considerations. Immersive experiences can themselves cause harm, from cybersickness to the distress of confronting feared stimuli during exposure, and the included studies rarely reported how such effects were monitored or managed. Many of the applications also involved vulnerable groups, including children and patients with dementia, stroke or serious mental illness, for whom comprehension, valid consent and the persuasive force of immersive content are especially pressing. Bias in the AI component compounds these risks, since a model trained on narrow data may perform unevenly across the very populations these systems are intended to serve.

5.5 Research Trends and Gaps

While **RQ**s **1–4** were addressed in Section 4, this section addresses the final, forward-looking research question **RQ5**. Two related trends shape the evidence base: increasing breadth and depth of AI-XR integration. In terms of breadth, after a stable volume from 2021 to 2024, study numbers rose sharply in 2025, and the work moved well beyond its imaging-heavy surgical origins into behavioural therapy, rehabilitation and supportive care [48, 66, 67]. In terms of depth, closed-loop designs, in which AI adapts the immersive experience in real time, were the most common mode, and recent work has increasingly combined several AI components or introduced generative and conversational elements instead of relying on a single classifier. These developments suggest growing depth in AI-XR integration, as the connection between AI and XR becomes more dynamic, multi-component, and embedded within clinical task environments.

Despite this momentum, the studies' reported limitations were consistent. Most studies were small, single-site, single-arm or non-randomised, with short or absent follow-up [36, 65]. Recurring technical and translational obstacles included hardware cost, registration error, latency, computational demand and manual setup, leaving several systems closer to proof-of-concept than workflow-ready tools [3, 43, 52]. Heterogeneous outcomes, positive-result skew and uneven coverage of diagnostics, pathology, public health and multidisciplinary use further limited synthesis.

5.6 Implications for Future Research

These gaps point to a clear research agenda. The field needs larger, controlled, and externally validated studies that compare integrated AI-XR systems against meaningful alternatives and report clinical outcomes rather than technical metrics alone. Studies should also report workflow behaviour, set-up burden, clinician acceptance, cost and sustained use. Established reporting standards for early-stage clinical AI [65] and digital-health evidence [64] would make studies more comparable and temper positive-result bias.

Opportunities exist to broaden both the technologies and clinical questions investigated. Generative or conversational AI is likely to become more common and should therefore be carefully evaluated. The under-represented domains of diagnostics, pathology and public health, along with multidisciplinary use, are targets for future research. As integration mode varies by clinical task, future systems should be designed around the task and workflow from the outset, rather than adding immersion to an existing AI system and vice versa. Standardised, patient-centred outcomes, cost reporting and clearer disclosure of AI training and validation data would further support comparison, adoption decisions and reproducibility. These priorities would help establish when, where, and how integrated systems provide meaningful value in clinical practice.

5.7 Strengths and Limitations

This review has several strengths. It used a systematic and PRISMA-guided approach across three major databases spanning biomedical, clinical, and technical research. The review spanned the full range of immersive modalities and a broad set of clinical domains rather than a single specialty, and coded each system at two levels: the AI method and the clinical function it served. This kept the technical and clinical readings distinct. The integration taxonomy provided a consistent way to compare otherwise heterogeneous systems, and the synthesis deliberately avoided equating positive reports with proven benefit.

This review also has some limitations. Several coding decisions, particularly integration mode, evidence maturity, and technology readiness level, required judgement. Although these decisions were documented to support transparency, some subjectivity remains. The heterogeneity of study designs, outcomes, and sample units, together with the strong predominance of positive findings, made meta-analysis inappropriate and limited the synthesis to structured narrative and visual approaches. The search was also restricted to three databases, peer-reviewed English-language publications, and the available evidence up to the search date, which may have excluded relevant studies. These constraints limit the generalisability of the findings and are themselves indicative of the early and heterogeneous state of the AI-XR evidence base.

6 Conclusion

This PRISMA-guided systematic review of 41 clinically relevant empirical studies (manually screened from 433 query matches) set out to clarify where and how the integration of AI with AR, VR and MR is approaching real-world impact in healthcare. Three findings stand out. First, AI-XR integration has moved beyond proof-of-principle, and is now being evaluated with patients, clinicians and clinical data across a widening set of applications. Second, the way the two technologies connect is governed by the clinical task, not by the AI method family or the display hardware, so the umbrella label AI-XR integration in fact covers structurally distinct architectures. The integration-mode taxonomy used in this review therefore provides a more meaningful basis for comparison than technology-centred groupings. Third, research activity should not be mistaken for clinical and technology maturity. Reported outcomes were almost uniformly positive, but the evidence concentrated at small-scale piloting, and no study reached testing at scale or routine adoption.

The implications are clear. Future work should move beyond feasibility through larger, controlled and externally validated studies that report clinical rather than purely technical outcomes, and that treat sustained use within real workflows as an outcome in its own right. Greater transparency about AI components and their training data, attention to under-represented modalities and domains, and clearer arrangements for consent, accountability and equitable access will all be needed. AI-XR is trending towards clinical impact in a few well-suited areas, but closing the gap between promise and durable, deployable benefit—rather than expanding the catalogue of pilots—is the greatest challenge for the next phase of this field.

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